

An IoT-Based Hybrid Fire-Fighting Robot with Real-Time Monitoring and Control System via a Web-Based Dashboard

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Abstract—Fire accidents pose a significant threat to human life and property, often requiring rapid response in hazardous environments where human intervention may be delayed or impossible. This paper presents the design and implementation of a WiFi-controlled autonomous fire fighting robot capable of detecting and extinguishing fire without human intervention. The system is built around two ESP32 microcontrollers: a main controller responsible for sensor processing, motor control, pump activation, and hosting a comprehensive web-based dashboard; and an ESP32-CAM module providing live video streaming with pan-tilt camera control and adjustable flash illumination. Three analog flame sensors enable directional fire detection, while an MQ2 gas sensor monitors smoke levels and a DHT11 sensor tracks ambient temperature and humidity. Upon fire detection, the robot autonomously navigates toward the flame, activates a water pump, and performs a servo-driven nozzle sweep to extinguish the fire. The web dashboard additionally supports full manual override, allowing independent control of robot movement, pump, nozzle servo, LEDs, and buzzer. Multicast DNS (mDNS) is employed for seamless device discovery without static IP configuration. Testing confirmed reliable fire detection, autonomous navigation, and effective fire suppression within a controlled environment.

Index Terms—Fire Fighting Robot, Embedded System, IoT, Hybrid Control, Remote Monitoring, WebSocket, mDNS, Web Dashboard

I. INTRODUCTION

Fire is one of the most destructive natural and man-made disasters, causing thousands of casualties and billions of dollars in property damage annually. Conventional fire response systems rely entirely on human intervention, which introduces dangerous delays and exposes firefighters to life-threatening conditions. According to the Bangladesh Fire Service and Civil Defence, Bangladesh recorded 26,659 fire incidents in 2024, resulting in 140 fatalities and 341 injuries. In 2025, 27,059 fire incidents were reported, causing 85 deaths and 267 injuries nationwide. These statistics highlight the urgent need for

faster, safer, and more intelligent fire detection and firefighting systems. Recent advancements in robotics, embedded systems, and Internet of Things (IoT) technologies have enabled the development of autonomous and remotely operated fire-fighting robots capable of reducing human involvement in dangerous rescue operations. Smart robotic systems can improve emergency response efficiency by detecting fire sources, monitoring environmental conditions, and performing extinguishing tasks in hazardous areas where human access is difficult or unsafe.

A. Motivation and Problem Statement

Traditional firefighting operations often require firefighters to enter hazardous environments filled with smoke, toxic gases, and extreme temperatures. In many emergency situations, delayed fire detection and limited accessibility significantly increase the risk of casualties and property damage. Manual firefighting methods are not always efficient in dangerous areas where immediate human intervention becomes risky. The motivation behind this research is to develop a smart fire-fighting robotic system capable of detecting fire and smoke automatically while providing real-time surveillance and remote control capabilities. Existing low-cost fire-fighting robots are often limited in functionality, lacking features such as live video streaming, multi-directional fire detection, environmental monitoring, and wireless web-based operation.

B. Project Objective

This project develops a dual-microcontroller fire-fighting robot that combines fully autonomous operation with a comprehensive WiFi-based control interface accessible from any browser on the local network. The proposed system is designed to integrate autonomous fire detection, real-time monitoring,

and wireless remote operation within a unified robotic platform. The robot is designed to improve operational safety, minimize human exposure to dangerous environments, and provide efficient real-time monitoring during fire emergencies. The primary objectives of this project are as follows:

- To detect fire directionally using three analog flame sensors and respond autonomously with appropriate navigation.
- To monitor smoke levels using an MQ2 gas sensor for early hazard detection.
- To provide real-time temperature and humidity monitoring.
- To extinguish fire using a water pump equipped with a servo-driven sweeping nozzle.
- To provide real-time live video monitoring through an ESP32-CAM module with pan-tilt servo control and adjustable flash illumination.
- To implement a responsive web based dashboard supporting both real-time sensor monitoring and full manual control of all actuators including motors, pump, nozzle servo, LEDs, and buzzer.
- To enable seamless multi-device discovery using mDNS without requiring static IP configuration.
- To improve operational safety by minimizing human exposure to hazardous fire environments.

C. Scope

The scope of this project includes the development of a prototype-level intelligent fire-fighting robot capable of operating in small-scale indoor environments. The system supports autonomous fire detection, wireless surveillance, environmental monitoring, and remote manual control through a web interface. The proposed robot can be applied in smart homes, laboratories, warehouses, industrial safety systems, and academic research related to robotics and IoT-based automation.

D. Related Work

Several researchers have developed automated systems to enhance fire safety by integrating microcontrollers, sensors, and communication technologies. An IoT-enabled hybrid firefighting robot was developed for industrial environments using multi-sensor fire detection, autonomous control, and 360° visual monitoring. The robot employs both water and AFFF foam suppression systems to achieve efficient fire detection, localization, and extinguishing with minimal human intervention [2]. An ESP32-based IoT firefighting robot was developed for real-time fire detection, affected area monitoring, and real-time information sharing. The robot can autonomously approach fire sources and facilitate quick response through integrated automation and remote-control capabilities [1]. A remotely controlled firefighting robot was developed using an ESP32-CAM module to provide real-time video streaming, fire detection, and water-based suppression. The robot supports remote navigation and monitoring, improving firefighting efficiency and situational awareness in emergency conditions [3].

II. SYSTEM DESIGN AND METHODOLOGY

A. System Architecture

The system consists of two ESP32 microcontrollers operating on the same WiFi network. The Main ESP32 serves as the primary controller, hosting a synchronous WebServer on port 80 and managing all robot functions. The ESP32-CAM module operates as a secondary controller, hosting an AsyncWebServer with two dedicated WebSocket endpoints: `/Camera` for binary JPEG frame streaming and `/CarInput` for pan, tilt, and flash control commands. Both devices register on the local network using mDNS — the main board as `firerobot.local` and the camera module as `firecam.local` — enabling IP-independent communication. The main ESP32 resolves the CAM IP at boot and embeds the ESP32-CAM page as an `iframe` within the main dashboard. If the CAM is not found initially, a retry occurs every 30 seconds automatically.

The overall system block diagram is shown in Fig. 1.

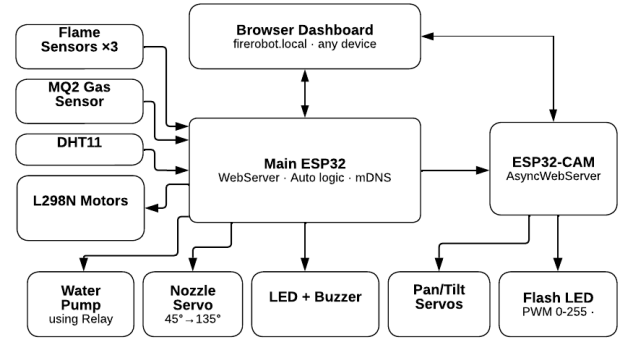


Fig. 1. System block diagram showing Main ESP32, ESP32-CAM, sensors, actuators, and browser dashboard connections.

B. Communication Flow

The communication flow between system components is as follows:

- 1) The browser connects to `firerobot.local` and loads the main ESP32 dashboard.
- 2) The dashboard polls `/status` every 500 ms via HTTP fetch for live sensor and robot state updates.
- 3) The dashboard embeds the ESP32-CAM page as an `iframe`, providing live video and camera controls using pan-tilt within the same interface.
- 4) The ESP32-CAM streams JPEG frames via WebSocket binary messages to an HTML5 canvas in the browser.
- 5) Pan, tilt, and flash commands are sent from the browser to the ESP32-CAM via a second WebSocket (`/CarInput`) as text messages (e.g., `Pan, 90`).

C. Hardware Components

Table I lists all hardware components with GPIO pin assignments and functions.

TABLE I
HARDWARE COMPONENTS AND PIN ASSIGNMENTS

Component	GPIO	Function
<i>Main ESP32</i>		
Flame Sensor (L)	34	Fire detect – left
Flame Sensor (M)	32	Fire detect – middle
Flame Sensor (R)	35	Fire detect – right
MQ2 Gas Sensor	16	Smoke/gas detection
DHT11	4	Temperature & humidity
L298N IN1/IN2	27/26	Left motor direction
L298N IN3/IN4	25/33	Right motor direction
L298N ENA/ENB	14/15	Motor speed (PWM)
Water Pump (Relay)	12	Fire extinguishing
Nozzle Servo	13	Spray sweep 45–135°
Red LED	18	Fire / danger alert
Green LED	19	Safe status indicator
Buzzer	17	Audio alert
<i>ESP32-CAM (AI-Thinker)</i>		
Pan Servo	12	Horizontal camera turn
Tilt Servo	13	Vertical camera turn
Flash LED	4	PWM brightness 0–255

D. Software Components

Implementation relies on:

- ESP32 Arduino Core for sensor interfacing, actuator control, and overall system management.
- Analog/digital I/O functions for reading data from flame sensors, MQ-2 gas sensor, and DHT11 sensor.
- PWM-based motor control routines for smooth directional navigation.
- Serial Monitor for debugging and data display.
- Custom logic functions for fire-direction detection and pump activation.
- Arduino framework with WiFi, WebServer, ESPmDNS, ESP32Servo, and DHT libraries for the main ESP32 controller.
- AsyncTCP, ESPAsyncWebServer, ESP32Servo, and the native esp_camera driver for the ESP32-CAM module.
- A responsive web dashboard implemented using plain HTML, CSS, and JavaScript embedded directly into the firmware as raw string literals.

E. Working Principle and Data Flow

1) *Autonomous Fire Fighting Mode*: The main control loop executes every 20 ms. Analog values from the three flame sensors are read on each iteration. A value below the threshold of 500 indicates fire in that direction. The autonomous flowchart is shown in Fig. 2.

The movement decision logic handles five cases:

- **Middle + Right**: Robot moves backward, then turns slightly right to reposition.
- **Middle + Left**: Robot moves backward, then turns slightly left.
- **Middle only**: Robot moves directly forward.
- **Left only**: Robot moves backward, then turns left.

- **Right only**: Robot moves backward, then turns right.
- **No fire**: Robot stops; pump off; green LED on.

Once positioned, the water pump activates and the nozzle servo sweeps from 45° to 135° and back to 90°, distributing water across the fire area. During the sweep, `handleClient()` is called every 10 ms to maintain dashboard responsiveness. After extinguishing, `resetRobot()` backs the robot away, switches off the pump, restores the green LED, and resets the servo to 90°. The DHT11 is sampled every 2 seconds independently.

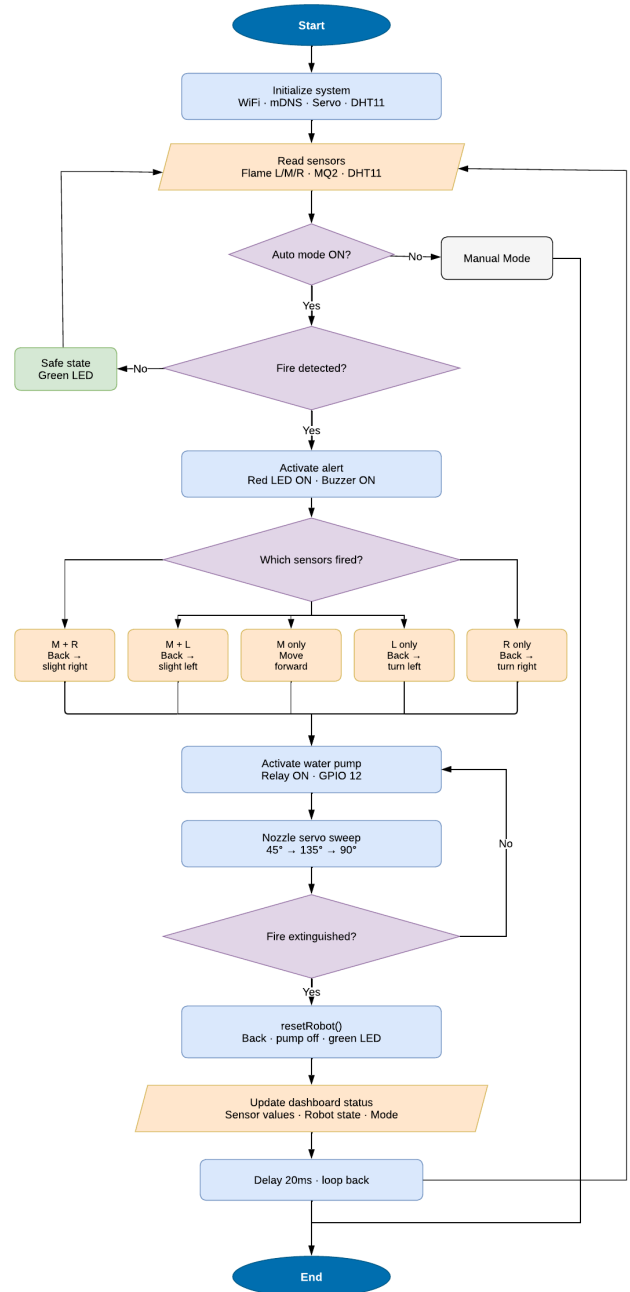


Fig. 2. Autonomous fire fighting flowchart using standard symbols: oval = start/end, rectangle = process, diamond = decision, parallelogram = input/output.

2) *Manual Control Mode*: Switching to Manual Mode disables all autonomous logic. The operator gains full individual control over every actuator:

- **Robot movement**: Forward, backward, left, right via D-pad with press-and-hold (100 ms repeat; stop on release).
- **Water pump**: Independent ON/OFF buttons.
- **Nozzle servo**: Slider (0°–180°) for precise manual direction.
- **Red LED**: Independent ON/OFF toggle.
- **Green LED**: Independent ON/OFF toggle.
- **Buzzer**: Independent ON/OFF toggle.

3) *Camera Control via ESP32-CAM*: The ESP32-CAM panel, embedded as an iframe, provides:

- **Live video**: VGA JPEG frames (640×480) via binary WebSocket, rendered on an HTML5 canvas.
- **Pan**: Horizontal servo slider (0°–180°) via WebSocket message `Pan,value`.
- **Tilt**: Vertical servo slider (0°–180°) via WebSocket message `Tilt,value`.
- **Flash**: PWM brightness slider (0–255) via WebSocket message `Light,value`.
- **Center reset**: Returns pan and tilt to 90°.

On disconnect, the ESP32-CAM resets pan/tilt to center and turns off flash automatically.

5) *Camera and Controls*: The ESP32-CAM streamed VGA frames with latency under 200 ms. Pan/tilt responded in under 100 ms. Flash PWM provided smooth brightness across the full 0–255 range. The robot is designed to improve operational safety, minimize human exposure to dangerous environments, and provide efficient real-time monitoring during fire emergencies.

C. Performance Summary

Table II summarizes key performance results.

TABLE II
KEY PERFORMANCE RESULTS

Parameter	Result
Fire detection range	20–30 cm
Flame sensor threshold	500 (analog)
Ambient sensor reading	>2000
Fire sensor reading	<300
Navigation accuracy	5/5 test cases
Nozzle sweep duration	≈2.16 s
Camera resolution	VGA (640×480)
Camera stream latency	<200 ms
Pan/Tilt response	<100 ms
Flash PWM resolution	8-bit (0–255)
Dashboard refresh	500 ms
mDNS compatibility	Windows & Android

III. IMPLEMENTATION AND RESULTS

A. System Setup

Both ESP32 boards were programmed using Arduino IDE 2.3.4 with the ESP32 board package. Both connect to the same WiFi network at startup. The main ESP32 registers as `firerobot.local` and queries for `firecam.local` after a 3-second boot delay. The dashboard is accessed at `http://firerobot.local` from any browser on the same network (tested on Windows and Android).

B. Testing and Results

1) *Fire Detection*: A lighter flame was used as the fire source. Flame sensors reliably detected fire at 20–30 cm. Ambient analog readings were above 2000; fire readings dropped below 300 against a threshold of 500.

2) *Autonomous Navigation*: Five directional scenarios were tested: left-only, right-only, middle-only, middle+left, and middle+right. The robot correctly identified the direction and executed the appropriate movement in all five cases.

3) *Fire Extinguishing*: The nozzle servo completed the full sweep (45°→135°→90°) in approximately 2.16 s at 5° steps with 60 ms per step. The pump remained active throughout.

4) *Manual Control*: All manual controls were verified: D-pad movement, pump toggle, nozzle servo slider, Red/Green LED toggles, and buzzer toggle. Mode switching between AUTO and MANUAL was instantaneous.

D. Prototype

Fig. 3 shows the assembled prototype. Fig. 4 shows the Web dashboard of live view, sensors & auto fire detection mode. Fig. 5 shows the Web dashboard of manual robot operation, camera control & monitoring. Fig. 6 shows the circuit wiring.

IV. RESULTS AND DISCUSSION

A. Functional Outcomes

The developed prototype successfully demonstrated the core functionalities of an intelligent WiFi-based fire-fighting robot using dual ESP32 microcontrollers and integrated sensing modules. The system was capable of detecting fire directionally using three analog flame sensors while simultaneously monitoring smoke levels through the MQ-2 gas sensor and environmental conditions using the DHT11 sensor. During testing, all sensor readings were displayed in real time through the web dashboard and Serial Monitor. When fire was detected, the robot automatically activated the buzzer and warning LEDs, navigated toward the fire source, and activated the water pump with a servo-driven nozzle to extinguish the flame effectively. The ESP32-CAM module successfully provided real-time live video streaming with pan-tilt camera control and adjustable flash illumination, enabling remote monitoring through a web browser. The responsive web dashboard also allowed manual control of robot movement, pump operation, nozzle positioning, LEDs, buzzer activation, and camera orientation. Overall, the system demonstrated reliable fire detection, stable wireless communication, smooth motor control, and effective fire-extinguishing performance under controlled testing conditions.

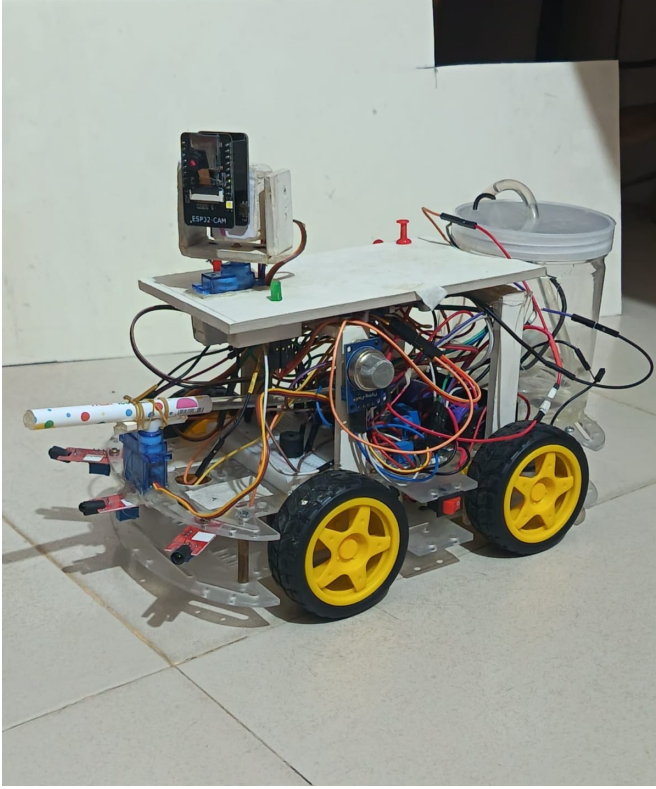


Fig. 3. Assembled fire fighting robot prototype.

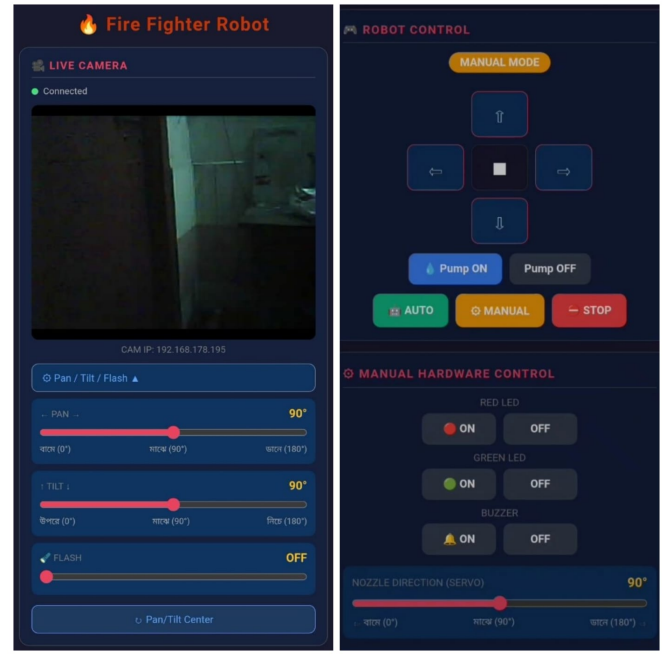


Fig. 5. Web dashboard: manual robot operation, camera control & monitoring.

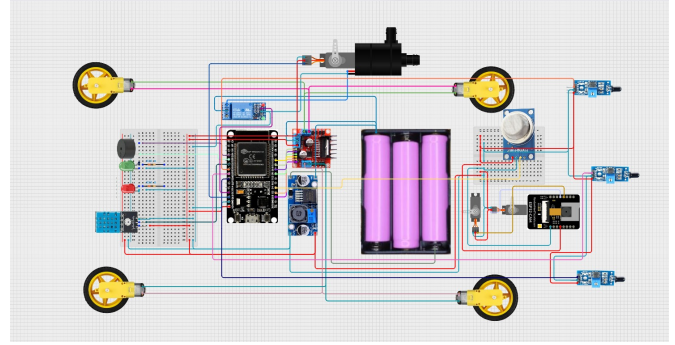


Fig. 6. Circuit diagram of the fire fighting robot.

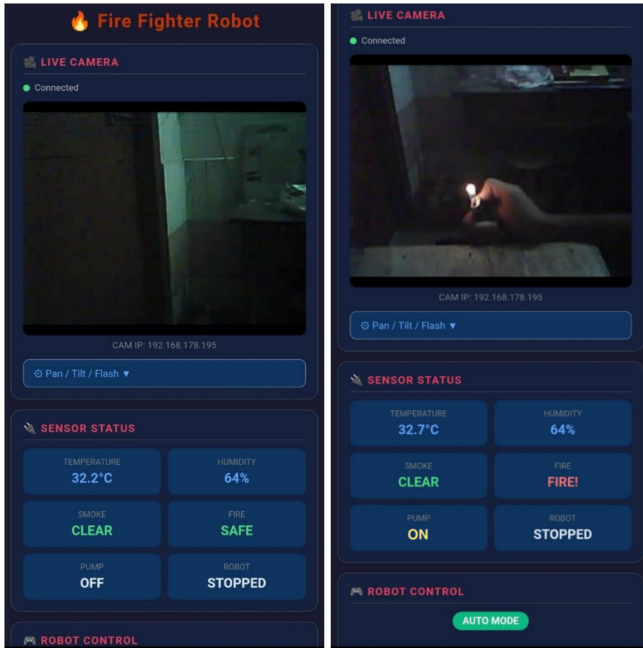


Fig. 4. Web dashboard: live view, sensors & auto fire detection mode.

B. Limitations and Challenges

Although the prototype performed successfully in most test scenarios, several limitations and technical challenges were encountered during development and testing:

- Simultaneous operation of the motors, water pump, ESP32 modules, and sensors occasionally caused voltage drops, affecting system stability and performance.
- False detections can occur when strong ambient light, reflections, smoke density variations, or environmental interference mimic fire signatures. To minimize this careful sensor calibration and threshold adjustment were implemented using combined flame, smoke, and temperature sensor readings, which reduce spurious triggers during experiments.
- Since the web dashboard and live video streaming depend on WiFi communication, unstable network conditions could affect real-time monitoring and responsiveness.

These limitations highlight the need for further optimization and system refinement to improve reliability, operational stability, and overall performance for real-world fire-fighting and safety applications.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

The proposed firefighter robot project successfully demonstrated the integration of sensors, actuators and micro-controller based control. The project successfully demonstrated autonomous fire detection, real-time live video streaming, environmental monitoring, wireless communication, and remote surveillance using dual ESP32 micro-controllers.. The project also provided hands-on experience in circuit design, sensor calibration and programming for real-time embedded systems. Lessons learned highlight the importance of systematic testing of individual components before full integration, effective troubleshooting techniques and the value of iterative design for improving overall system performance. Overall, the project met its objectives of building a functional prototype capable of navigating its environment and responding to fire sources effectively.

B. Social Relevance

The proposed firefighting robot is socially significant because it addresses the critical need for early fire detection, real-time monitoring, and rapid emergency response, helping to reduce property damage and save lives. By combining automatic fire detection and response with manual control of both robot movement and the fire extinguishing system, it enables safe and effective operation in hazardous environments. The integration of real-time sensor monitoring, live video streaming, and a pan-tilt camera enhances situational awareness, making the system suitable for homes, offices, and industrial facilities while contributing to overall safety and emergency preparedness.

C. Future Work

In the future, several improvements can be implemented in the future to enhance the system's intelligence, reliability, and overall performance:

- 1) Integrate computer vision and AI-based fire classification using the ESP32-CAM module to identify different fire types from flame by analyzing fire color and visual patterns (chemical or electrical fires), helping avoid inappropriate extinguishing methods in hazardous situations.
- 2) Integrate advanced multi-gas sensing technologies capable of identifying specific hazardous gases present in smoke, enabling more accurate environmental monitoring and improving safety during fire emergency situations.
- 3) Implement obstacle detection and autonomous navigation features for safer movement in complex environments.
- 4) Integrate cloud-based IoT platforms such as Firebase for remote monitoring and instant mobile notifications.
- 5) Improve power management using dedicated voltage regulation and battery monitoring systems.

- 6) Extend live video streaming capabilities for internet-based remote access beyond the local network.
- 7) Improve the water spraying mechanism with adaptive nozzle positioning and pressure control.
- 8) Optimize the robot chassis and motor system for operation in more complex environments.

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REFERENCES

- [1] M. A. Hossain, H. S. Roy, M. F. K. Khondakar, M. H. Sarowar, and M. A. Hossainline, "Design and Implementation of an IoT Based Firefighting and Affected Area Monitoring Robot," in *2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, 2021, pp. 552–556, doi: 10.1109/ICREST51555.2021.9331064.
- [2] M. S. Al Basar and R. Roy, "An IoT Based Smart Hybrid Firefighting Robot with 360° Visual Intelligence for Industrial Safety Applications," in *2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025, pp. 1–6, doi: 10.1109/COMPAS67506.2025.11381863.
- [3] A. Yahya, A. Sarhan, L. Omar, M. Derar, B. Al-Qadi, S. Alsadi, T. Foqha, L. A. Maghrabi, and M. Kanan, "Remotely Controlled Firefighting Robot with ESP32-CAM Module," in *2023 2nd International Engineering Conference on Electrical, Energy, and Artificial Intelligence (EICEEI)*, 2023, pp. 1–4, doi: 10.1109/EICEEI60672.2023.10590289.
- [4] Espressif Systems, "ESP32 Series Datasheet," Espressif Systems, Shanghai, China, 2023. [Online]. Available: https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf
- [5] me-no-dev, "ESPAsyncWebServer Library," GitHub Repository, 2023. [Online]. Available: <https://github.com/me-no-dev/ESPAsyncWebServer>
- [6] Espressif Systems, "mDNS Service," ESP-IDF Programming Guide, 2024. [Online]. Available: <https://docs.espressif.com/projects/esp-idf/en/latest/esp32/api-reference/protocols/mdns.html> (Accessed: Aug. 20, 2025).
- [7] D-Robotics, "DHT11 Humidity and Temperature Sensor Datasheet," D-Robotics UK, 2010.
- [8] Zhengzhou Winsen Electronics Technology Co., Ltd., "MQ-2 Semiconductor Sensor for Combustible Gas Datasheet," Winsen Electronics, 2014.
- [9] R. Santos and S. Santos, "ESP32-CAM," *Random Nerd Tutorials*, 2023. [Online]. Available: <https://randomnerdtutorials.com/esp32-cam-video-streaming-web-server-camera-url/>